

Mirror cleaning



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1 General

1.1 Manufacturer address

ASA Astrosysteme GmbH

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AUSTRIA

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Email: office@astrosysteme.com

Homepage: <https://www.astrosysteme.com>



1.2 Support

If you experience problems or have further technical questions feel free to contact our support at support@astrosysteme.com.

If you have questions concerning accessories or prices consult our sales department at sales@astrosysteme.com.

1.3 Changelog

Version Nr	Date	Change
A	2022-09	Release first version

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2 User information

	Caution against optical radiation;	Eyepiece-/camera mounting point	Do not look into the telescope when the telescope is aiming on a bright object. The bright light might harm your eyes.
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General

The following instructions apply to cleaning the mirrors in the installed state.

Process:

Make mirrors accessible;

Cleaning;

Restore operational readiness;

2.1 Required materials

Detergent	2 spray bottles
Distilled water	Kitchen roll (paper cloths)
Isopropanol	Microfiber cloth

	You can leave the mirror inside the mirror cell for cleaning. The mirrors do not have to be removed.
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	Do never wipe on the dry mirror surface! Do never apply pressure when wiping on the mirror surface!
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3 Tertiary mirror (M3)

The following instructions describe how to clean a tertiary mirror.

3.1 Make the mirror accessible

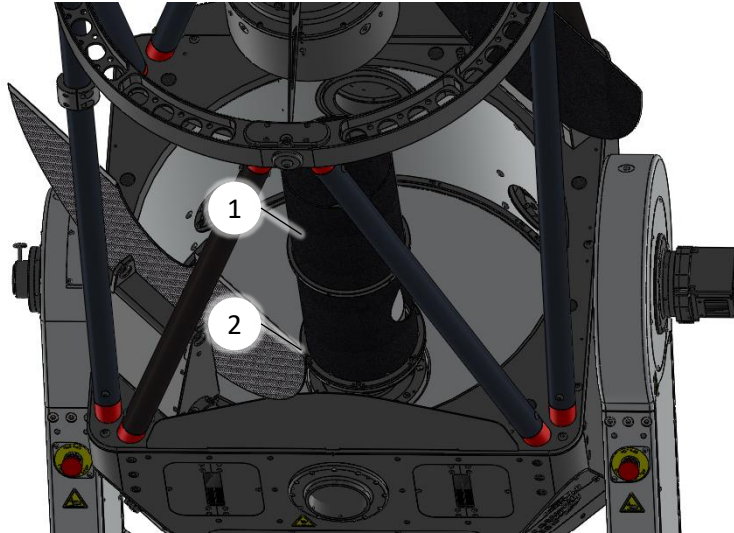


Figure 1: remove baffle M3

1. Deactivate the device.
2. Put the elevation axis in a position where all liquids can flow off the mirror.
3. Fix the elevation axis with the brake.
4. Remove the baffle (1) to gain access to the mirror. Therefore remove the screws (2) of the baffle and pull it up.

Be careful not to touch the tertiary mirror!



5. Put some kitchen paper (1) around the edge of the mirror to gather flowing-off liquids.

3.2 Cleaning



1. If available, use compressed air (oil free) to remove dust particles from the mirror surface.
2. Put distilled water into the spray bottle.
3. Wet the mirror with distilled water



4. Cover the mirror with paper cloths (1). Wet them with distilled water. They should be wet and stick on the mirror surface.
5. Leave them on the mirror for a few minutes.



6. Remove the paper from the mirror by just pulling it off. Do not wipe yet. With this procedure you do not produce any scratches.
7. Repeat the procedure 2-3 times.



8. Put distilled water and detergent into one of the spray bottles.
9. Wet the mirror with this mixture.



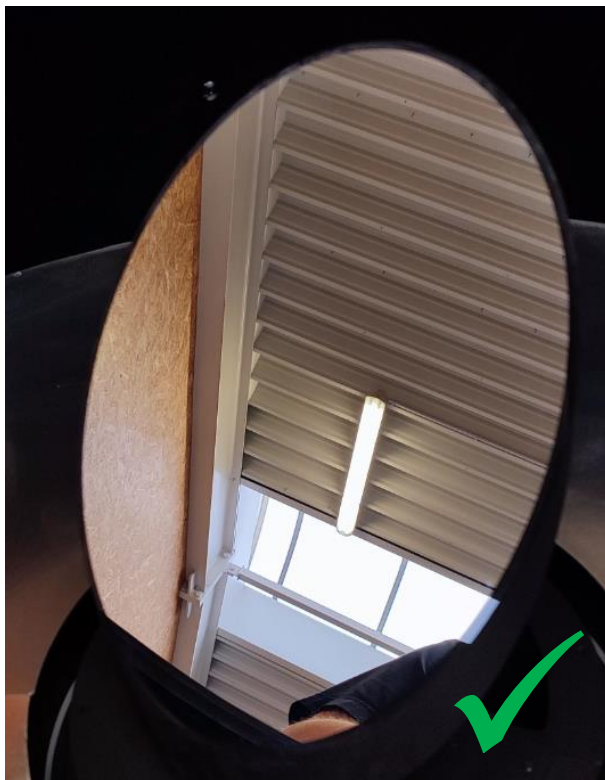
10. Wipe with the microfiber cloth (1) but without any pressure and keeping it completely wet. Only let the weight of the cloth touch the mirror, DON'T PRESS.
11. Repeat the procedure 2-3 times.



12. Put isopropanol into one of the spray bottles.
13. Wet the mirror with it.
14. The alcohol flows down the mirror and takes the water with it and dries up without leaving droplets like water does.



15. Wipe with the microfiber cloth (1) but without any pressure and keeping it completely wet. Only let the weight of the cloth touch the mirror, DON'T PRESS.
16. Repeat the procedure 2-3 times.
17. Let the mirror dry.



18. If streaks are still visible on the surface of the mirror, proceed as follows:
19. Breathe on the mirror;
20. Wipe with the microfiber cloth (1) but without any pressure and keeping it completely wet. Only let the weight of the cloth touch the mirror, DON'T PRESS.



If there is still some residual dirt on the mirror, just leave it there. Most scratches occur due to too intense mirror cleaning. Especially wiping on the dry mirror surface produces scratches.

3.3 Restore operational readiness

1. Remove all cleaning material.
2. Let the telescope dry.
3. Reassemble everything.

4 Primary Mirror (M1)

4.1 Make the mirror accessible

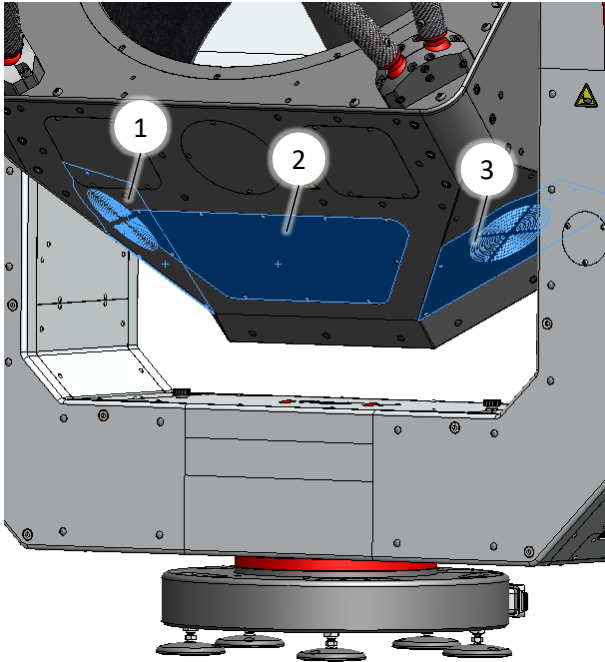


Figure 2: AZ telescope

1. Deactivate the device.
2. Remove covers (1, 2, 3) to gain access to the primary mirror.
3. Put some kitchen paper around the edge of the primary mirror to gather flowing-off liquids.
4. Put the elevation axis in a horizontal position and fix it with the brake.
5. Now, the mirror is in a vertical position so that all liquids can flow off the mirror.

4.2 Cleaning

Workflow see chapter 3.2.

4.3 Restore operational readiness

1. Remove all cleaning material.
2. Let the telescope dry.
3. Reassemble everything.

5 Secondary mirror (M2)

5.1 Make the mirror accessible

1. Deactivate the device.
2. Put the elevation axis in a horizontal position and fix it with the brake.
3. Put some kitchen paper around the edge of the secondary mirror and baffle to gather flowing-off liquids.

5.2 Cleaning

Workflow see chapter 3.2.

5.3 Restore operational readiness

1. Remove all cleaning material.
2. Let the telescope dry.
3. Reassemble everything.